

[CASE REPORT]

Immunoglobulin G4-related Disease with Solitary Nasopharynx Involvement Mimicking Nasopharyngeal Tumor: A Case Report

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Abstract:

Immunoglobulin G4-related disease (IgG4-RD) presenting with isolated nasopharyngeal involvement is rare. A 51-year-old man presented with left-sided otalgia and temporal tingling. Magnetic resonance imaging showed a 2.5-cm infiltrative soft tissue lesion in the left nasopharynx, initially suspected of being nasopharyngeal carcinoma. Serum IgG4 levels were normal. The first biopsy was inconclusive; however, a second, deeper biopsy confirmed IgG4-RD based on storiform fibrosis, dense lymphoplasmacytic infiltration, and increased IgG4-positive plasma cells. The patient responded well to moderate-dose glucocorticoids and azathioprine. IgG4-RD should be considered in the differential diagnosis of mass-forming lesions in the nasal cavity, even in the absence of elevated serum IgG4 levels.

Key words: Immunoglobulin G4-related disease, nasopharyngeal tumor, diagnostic challenge

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Introduction

Immunoglobulin G4-related disease (IgG4-RD) is an immune-mediated fibroinflammatory condition that can affect multiple organs and results in the formation of tumor-like masses. Characteristically, IgG4-RD presents with organ enlargement mimicking neoplasms and frequently involves the exocrine glands, pancreas, and retroperitoneum. This tendency to form mass-like lesions, such as dacryoadenitis, retroperitoneal fibrosis encasing the abdominal aorta, or diffuse pancreatic enlargement, can cause it to be mistaken for malignancies like pancreatic adenocarcinoma (1). While most patients have elevated serum IgG4 (sIgG4) levels, diagnosing IgG4-RD remains challenging due to its diverse clinical manifestations and ability to affect various organs. The diagnostic difficulty is further compounded when IgG4-RD involves less commonly affected organs or presents with normal sIgG4 levels. Despite the availability of classification criteria (2), an accurate diagnosis in such atypical scenarios

remains difficult.

We herein report a rare case of IgG4-RD with solitary nasopharyngeal involvement and normal serum IgG4 levels that clinically and radiologically mimicked a nasopharyngeal carcinoma.

Case Report

A 51-year-old man presented to the otolaryngology (ENT) department with a 3-month history of left-sided otalgia, a tingling sensation in the temporal area, and soreness. The patient was an ex-smoker with no significant medical history. He had no known history of allergic diseases, such as allergic rhinitis, bronchial asthma, eosinophilic sinusitis, or other atopic conditions, and reported no systemic symptoms. There was no family history of autoimmune diseases and no history of relevant occupational or environmental exposures. An endoscopic nasal evaluation revealed an irregular infiltrative lesion in the left nasopharynx. Magnetic resonance imaging (MRI) demonstrated a 2.5-cm mass-like infiltrative

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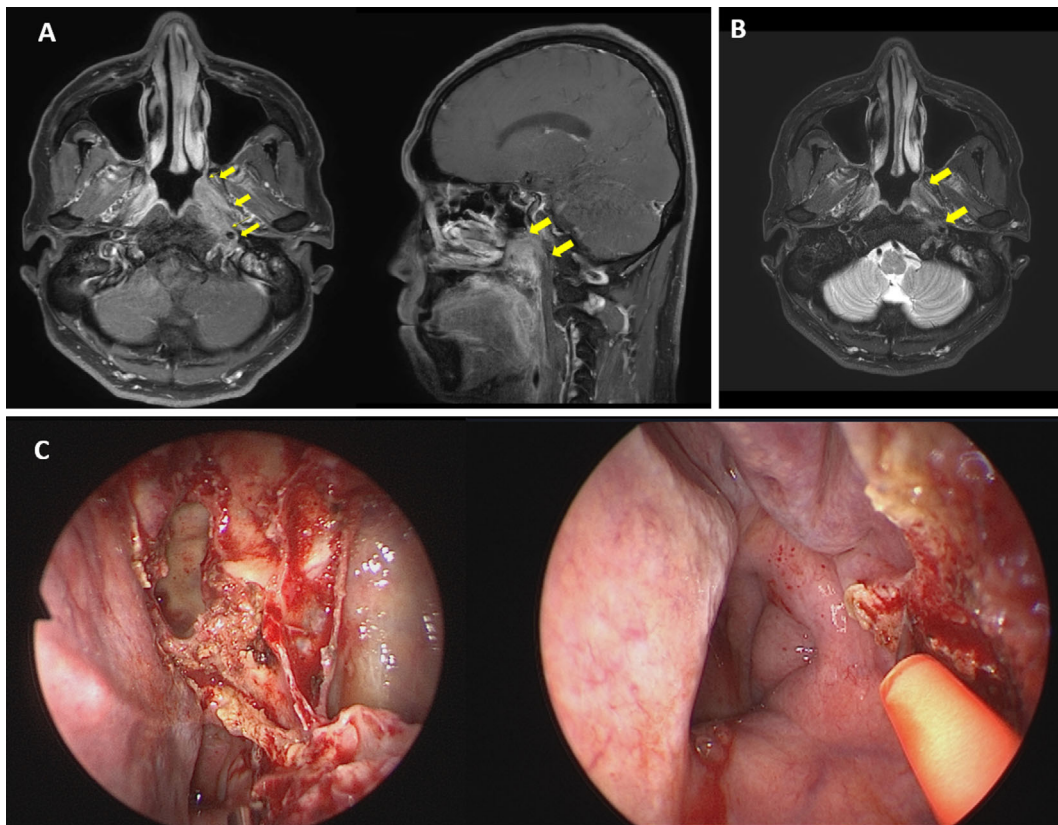


Figure 1. (A) Paranasal sinus (PNS) MRI showing infiltrative enhancing thickening of the left nasopharynx (yellow arrows) in the T1-weighted contrast-enhanced axial and sagittal view, (B) PNS MRI in T2-weighted axial view. (C) Nasal endoscopic evaluation revealing an irregular infiltrative lesion of the left nasopharynx.

soft tissue lesion in the left nasopharynx, accompanied by chronic otitis media in the left middle ear, suggesting nasopharyngeal cancer (Fig. 1).

The initial nasopharyngeal biopsy revealed lymphoplasmacytic and neutrophilic infiltration of some focal IgG4 positive plasma cells (up to 20/high-power field [HPF]). The patient was subsequently referred to the rheumatology department for a further evaluation of systemic inflammatory diseases. Laboratory investigations showed a mildly elevated erythrocyte sedimentation rate (ESR, 40 mm/h; normal range: <20 mm/h) and C-reactive protein level (CRP, 0.6 mg/dL; normal range: <0.5 mg/dL) with a normal eosinophil count of 192/mm³, negative anti-neutrophil cytoplasmic antibodies (ANCA), and normal serum IgE (51.6 U/mL; reference range: 0-100 U/mL), IgG (1,105 mg/dL; reference range: 700-1,700 mg/dL) and IgG4 (81.9 mg/dL; reference range: 3.9-86.4 mg/dL) levels. Chest and abdominal computed tomography (CT) revealed no other organ involvement.

To further investigate the possibility of other inflammatory conditions, such as granulomatosis with polyangiitis, a second deep biopsy was performed. Histopathology demonstrated fibrosis, dense lymphoplasmacytic and some eosinophilic infiltration with a significant increase in IgG4-positive plasma cells (80-90/HPF) (Fig. 2). IgG4-positive plasma cells were counted in three ×400 HPFs with the highest

number of IgG4-positive plasma cells and expressed as the average number of cells per HPF, in accordance with the 2019 American College of Rheumatology (ACR) and European Alliance of Associations for Rheumatology (EULAR) classification criteria (2, 3). Storiform fibrosis was defined as irregularly whorled or cartwheel-like patterns of fibrosis interspersed between inflammatory cell infiltrates.

Based on these findings, the patient was finally diagnosed with IgG4-RD limited to the nasopharynx. Treatment with moderate-dose glucocorticoids (methylprednisolone 0.5 mg/kg) and azathioprine led to symptomatic improvement. After nine months of therapy, follow-up MRI demonstrated complete resolution of the infiltrative enhancing lesion in the left side of the nasopharynx-parapharyngeal space-carotid space (Fig. 3).

Discussion

IgG4-RD is an insidiously progressive and often highly destructive fibroinflammatory disease characterized by chronic activation of the immune system and tissue fibrosis (4). Its autoimmune nature is supported by responsiveness to immunosuppressive therapy and association with autoantibodies, definitive pathogenic mechanisms remain unclear. IgG4-RD can affect nearly all organs but primarily targets the major salivary glands (submandibular, parotid,

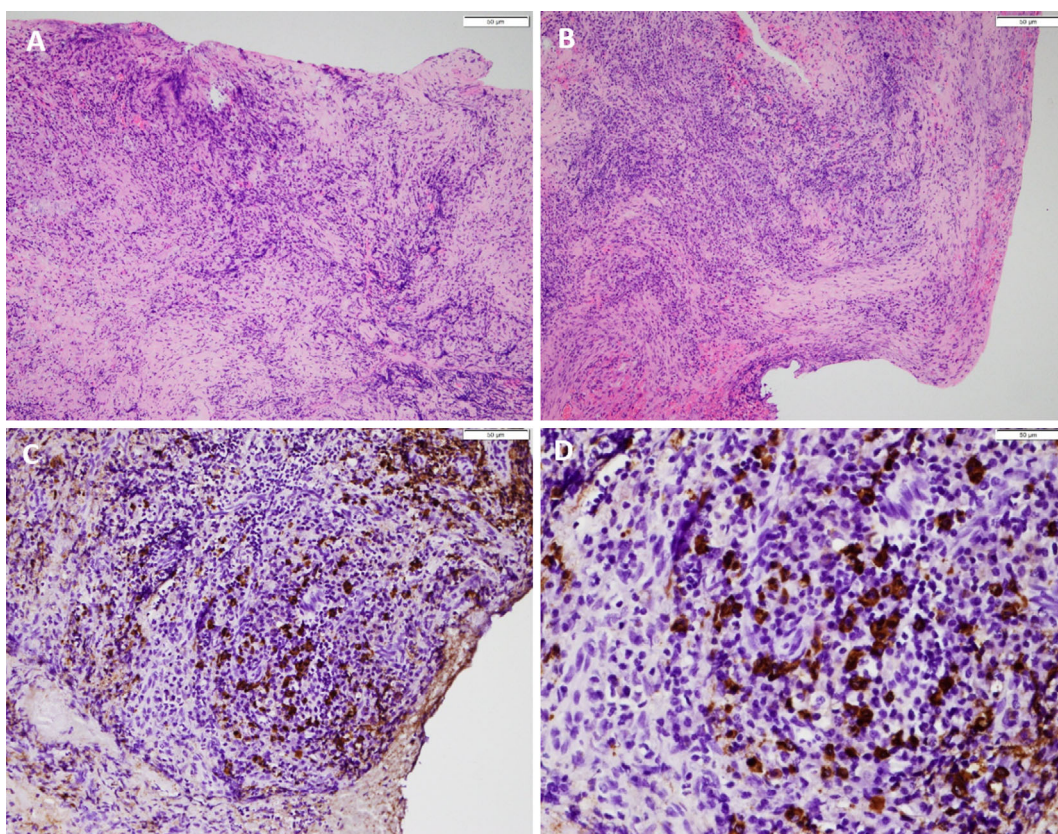


Figure 2. (A, B) Hematoxylin and Eosin staining demonstrating fibrosis with dense lymphoplasmacytic and some eosinophilic infiltration (original magnification $\times 100$). (C, D) Immunohistochemical staining for IgG4 showing abundant IgG4-positive plasma cells (80-90/HPF; C: $\times 200$, D: $\times 400$).

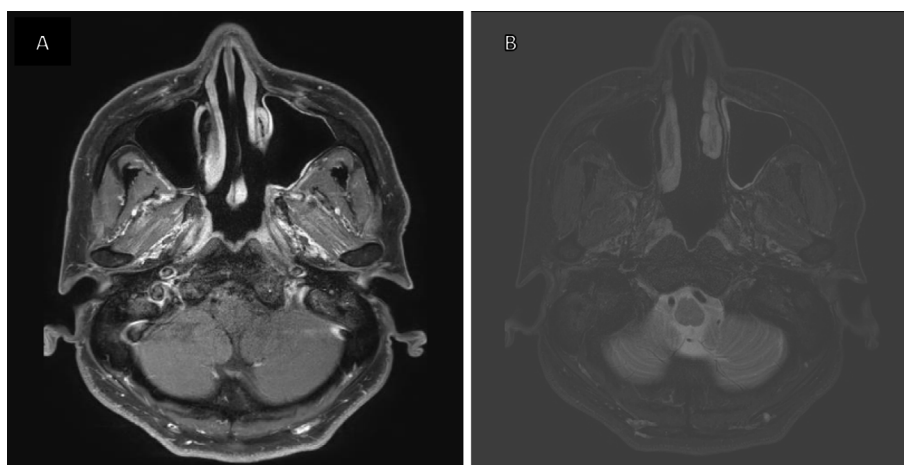


Figure 3. Follow-up PNS MRI showing completely resolved infiltrative enhancing lesion in the T1 (A) and T2 (B) -weighted-enhanced axial view.

and sublingual), orbits, lacrimal glands, pancreas, biliary tree, lungs, kidneys, aorta, retroperitoneum, meninges, and thyroid gland (Riedel thyroiditis) (5). Despite this wide range of involvement, organ-specific presentations can resemble malignancies or other inflammatory diseases, complicating the diagnosis.

The diagnosis of IgG4-RD is established through a combination of clinical features, increased serum IgG4 levels, and characteristic histopathological findings. However, ele-

vated serum IgG4 concentrations are neither highly sensitive nor specific. In addition, hallmark histopathological findings, namely storiform fibrosis, dense lymphoplasmacytic infiltration, and increased IgG4-positive plasma cells, are not exclusive to IgG4-RD. In 2019, the ACR and EULAR jointly endorsed the classification criteria for IgG4-RD (2). These criteria include head and neck gland involvement; however, they are limited to specific exocrine glands, such as the lacrimal and submandibular glands. Notably, nasopharyngeal

Table. Summary of Published Cases of Nasopharyngeal Involvement in Patients with IgG4-RD.

Author [ref.]	Age/ Sex	Symptoms	Site	Serum IgG4 Level (mg/dL, reference range)	Histopathological findings	Treatment	Outcome
Sasaki et al.[11]	71/M	nasal occlusion and swelling of the right cheek	nasal cavity and both maxillary sinuses	114 (4.8-105)	Immunostaining: plasma cells expressing IgG4	PD 40 → 10 mg/day 18mo	remission
Detiger et al.[12]	73/M	headache and hearing loss	left nasopharynx and clivus	110 (80–140)	IgG4-positive cells 380/HPF IgG4/IgG ratio 0.68	PD 40 mg/day. tapering for 11mo	remission
Liu et al. [13]	71/M	hearing loss, ringing in right ear, and upper neck pain	right nasopharynx skull base	165 (8–140)	IgG4-positive cells 20 to 30 /HPF IgG4/IgG ratio>0.4	MP 48 mg, followed by a tapering dose for the next 2 weeks. 1mo.	remission
Lin et al. [14]	63/M	Headache	nasopharynx	218	IgG4-positive cells 22/HPF.	MP 40mg qd CTX 0.2g qod	remission
	53/M	Epistaxis, Diplopia	nasopharynx	324	IgG4-positive cells >10/HPF.	MP 40mg qd CTX 50mg qd	remission
	64/F	Headache	nasopharynx	268	IgG4-positive cells 75/HPF.	MP 40mg qd CTX 50mg qd	remission
	67/M	Headache	nasopharynx	765	IgG4-positive cells >10/HPF.	MP 40mg qd CTX 50mg qd	remission
Thonpiya et al. [15]	50/F	left nasal obstruction, recurrent sinus infections, frequent epistaxis, muffled left ear	nasopharynx	NA	NA	high-dose PD with gradual tapering During a 3-month follow-up,	remission
Polianskis et al. [16]	31/F	right-sided facial weakness, headache, right ear hearing loss, and a slight decrease in right-sided facial sensitivity	right infratemporal fossa, nasopharynx, spreading intracranially to the middle cranial fossa with perineural extension	normal	IgG4-positive cells 80/HPF IgG4/IgG ratio 0.63	MP 500 mg for 3 days followed by a tapering dose (oral MP 16mg) 2mo. → after recurrence, prolonged steroid therapy without discontinuation at 16 months	total resolution 2mo. → at 7mo neurological symptoms recurred, but no pathologic infiltration on MRI
Ozsen et al. [17]	52/M	NA	nasopharynx	NA	IgG4-positive cells 12-14/HPF IgG4/IgG ratio 0.43	NA	NA
Ramanathan et al. [18]	75/M	headache, right aural block, tinnitus and hard of hearing	left nasopharynx	448 (uo to 200)	IgG4-positive cells 10-15/HPF IgG4 to IgG ratio >0.4	steroids.	NA
Present Case	51/M	left otalgia and temporal area tingling sensation and soreness	left nasopharynx	81.9 (3.9-86.4)	IgG4-positive cells 80-90/HPF	MP 0.5mg/kg azathioprine 9mo	remission

M: male, PD: prednisolone, mo: month, HPF: high-power field, MP: methylprednisolone, CTX;cyclophosphamide, NA;non available

involvement is not addressed in these criteria.

Nasopharyngeal involvement in IgG4-RD is a rare manifestation and can present with a variety of symptoms depending on the site of involvement. Reported symptoms range from rhinorrhea and nasal congestion to otologic manifestations such as tinnitus, serous otitis media, and hearing loss some of which were seen in our case, (6). When IgG4-RD involves the nasal cavity with perineural and bone marrow invasion, it can closely mimic a malignant tumor. Differential diagnoses include lymphoma, adenoid cystic carcinoma, squamous cell carcinoma, olfactory neuroblastoma, malignant melanoma, and other malignant tumors (7).

The differential diagnosis of ANCA-negative eosinophilic granulomatosis with polyangiitis (EGPA) was carefully considered in the present case due to the presence of focal eosinophilic infiltration in the tissue specimen. However, the patient had no history of allergic diseases, such as asthma, allergic rhinitis, eosinophilic sinusitis, or other atopic conditions. Tissue eosinophils accounted for <10% of inflammatory cells, which did not meet the criteria for eosinophil-predominant inflammation. The absolute eosinophil count was $<0.5 \times 10^9/L$, and there was no obstructive airway disease or nasal polyps, yielding no points under the 2022 ACR/EULAR classification criteria for EGPA (8). A histopathological examination revealed no necrotizing vasculitis or granu-

lomatous inflammation, and there were no systemic vasculitic manifestations. IgG4-RD has been associated with allergic conditions such as asthma, allergic rhinitis, and atopic dermatitis, and this overlap can complicate the differentiation from EGPA (9). In our patient, the serum IgE level was within the normal range at 51 U/mL. Although elevated serum IgG4 concentrations have been reported in some patients with EGPA and may correlate with disease activity (10), our patient had normal serum IgG4 levels and lacked clinical or laboratory features suggestive of EGPA. Taken together, these findings, combined with the characteristic histopathologic features, supported the diagnosis of IgG4-RD over EGPA in this case.

To date, only a few cases of solitary nasopharyngeal involvement in IgG4-RD have been published (Table) (11-18). Among these, only two cases with available data had normal serum IgG4 levels and no systemic involvement. Treatment outcomes were generally favorable, with most patients responding well to moderate-dose glucocorticoids. Only one patient achieved remission following high-dose glucocorticoid and radiation therapy. In one study, repeat biopsies were necessary to establish a definite diagnosis (12). Consistent with these findings, our patient also required a second, deeper biopsy to confirm the diagnosis. Compared with the previously reported cases that also presented with normal serum IgG4 levels, the case by Polianskis et al. (16) involved extensive spread from the infratemporal fossa and nasopharynx into the middle cranial fossa with perineural extension, requiring prolonged steroid therapy without discontinuation at 16 months. The case by Detiger et al. (12) involved the left nasopharynx and clivus, with potential cranial nerve risk. In contrast, our patient's disease was confined to the left nasopharynx without skull base or cranial nerve involvement, and sustained remission was achieved for nine months following glucocorticoid induction and azathioprine maintenance.

Although nasopharyngeal involvement is rare, sinonasal manifestations of IgG4-RD-including those affecting the nasal cavity, sinuses, and skull base-have been described in case series as part of the broader spectrum of head and neck involvement (19, 20). Furthermore, it is increasingly frequently recognized that patients with single-organ IgG4-RD are more likely to have normal serum IgG4 levels than those with multiorgan involvement (21), demonstrating the importance of histopathological evaluation. The present case contributes to this growing body of evidence, emphasizing the need for high clinical suspicion and adequate tissue sampling, even when serum IgG4 levels are within the normal range.

Due to its rarity, nasal cavity involvement is not captured in the 2019 ACR/EULAR classification criteria for IgG4-RD. Nonetheless, clinicians should consider IgG4-RD in the differential diagnosis of inflammatory mass lesions in the nasal cavity, particularly when histologic findings are suggestive and serum IgG4 level is within the normal range. A deep tissue biopsy can be essential for the diagnosis in cases

with such atypical presentations.

The authors state that they have no Conflict of Interest (COI).

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Informed consent: Informed consent was obtained from a patient included in the study.

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